

Ghifar Khder

Biomedical Engineer | M.Sc. Medical Systems Engineering student

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

Profile

Biomedical engineering graduate studying Medical Systems Engineering at OVGU Magdeburg. My project work includes medical image analysis, machine learning and biomedical electronics. I enjoy understanding how a system works, building it and testing it. I am interested in medical device development, particularly imaging and MRI technology.

Education

M.Sc. Medical Systems Engineering

Apr 2026 - present

Otto von Guericke University Magdeburg, Germany

Faculty of Electrical Engineering and Information Technology.

B.Sc. Biomedical Engineering

Sep 2018 - Aug 2023

Al Andalus University for Medical Sciences, Syria

Final grade: German equivalent 2.0. Ranked among the top 5 students in the graduating class.

Projects

Hardware projects · [View in portfolio](#)

Automatic CPR compression prototype

- DC motor and rack-and-pinion mechanism with selectable 4, 5 and 6 cm travel and adjustable compression rate.
- The entire design uses no microcontroller. Discrete control circuitry handles motor reversal and position sensing. An RC filter and NE555 pulse shaper address switch bounce at the endpoints.

Wearable pulse oximeter

- Wrist-worn prototype combining a MAX30100 sensor, ESP32, OLED display and rechargeable battery to display pulse rate and SpO₂.
- Includes browser-based monitoring over Wi-Fi, a rolling measurement log and CSV export. Compared prototype readings with a commercial fingertip oximeter.

Experimental optical glucose sensing prototype

- 940 nm infrared emitter and phototransistor with an LM358 amplifier, low-pass filter and Arduino readout.
- Explored a voltage-to-glucose calibration using paired home-meter readings.

Knee osteoarthritis grading

- Built a ResNet classifier for five knee osteoarthritis grades, using a conditional GAN to balance training data.
- Deployed a Streamlit app with image upload, grade prediction and Grad-CAM heatmaps. [GitHub](#)

Coronary stenosis detection

- Used ResNet101V2 to localise stenosis in about 8,300 coronary angiography images. Evaluated bounding boxes using IoU metrics and deployed a Streamlit app. [GitHub](#)

Diabetes prediction

- Combined five machine learning models with an ANN meta-learner on the Pima Indians Diabetes dataset. Included missing-value estimation and fuzzy clustering.
- Built an interface for records and CSV files, with model comparisons and predictions. [GitHub](#)

EEG sleep phase classification

- Trained a 1D CNN to classify two-second, single-channel EEG segments into CAP phases A and B. Built an interface to plot signals and compare predictions with reference labels. [GitHub](#)

Crop disease and yield prediction · [GitHub](#)

Experience

Founder and Team Leader

Jan 2022 - Apr 2026

On Time Team | Volunteer | Latakia, Syria

- Founded and coordinated a team supporting students with technical projects. Organised work, reviewed progress and acted as the main student contact.
- Designed hardware with the team and completed most programming projects independently.

Laboratory Receptionist and Workflow Coordinator

Jan 2023 - Mar 2024

Private laboratory | Latakia, Syria

- Managed patient records, enquiries, results and payments; coordinated with laboratory staff.

Department Lead

Jan 2022 - Jan 2023

Mad Matrix Team | Volunteer | Tartous, Syria

- Collected students' project ideas, discussed and planned them with students and team members, connected people and contributed to each stage of the team's projects.

Skills

Personal skills: Strong analytical thinking, a solid understanding of mathematics, and the ability to learn new concepts and tools very quickly.

Data processing and machine learning: Strong knowledge of data and image processing and analysis; deep learning with ANNs, CNNs and residual CNNs, machine learning and clustering algorithms, transfer learning, ensemble methods, and model evaluation principles.

Programming and tools: Python, GitHub, Streamlit, Microsoft Office.

Electronics: Circuit design, analysis and assembly, electrical measurements, analog amplification and filtering, sensor integration. Experience with Arduino and ESP32, and the ability to learn other microcontroller platforms very quickly.

Languages

Arabic: native | English: C1 | German: B1-B2